Optimization for Deep Learning (OPT4DL)

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Stochastic Lower Bounds

Motivation



- Recall our canonical training task

$$\min_w f_p(w) := \mathbb{E}_{z\sim p}[\ell(w,z)] \tag{1}$$

  for which we have discussed several efficient training methods, e.g., SGD and its variations
- Notably, we have discussed several upper bounds on the performance of these methods
- Does not tell us the fundamental limit of performance: best achievable by any method using stochastic gradients
- We will discuss tools to derive such results for stochastic convex optimization.
- Setup:
  - p is unknown, ℓ is convex and G-Lipschitz for all z
  - We observe stochastic gradients $g_1, \ldots, g_T$
  - We produce $\hat{w}$ as a function of $g_1, \ldots, g_T$

Minimax Optimality

- Recall, for SGD, from stochastic gradients $g_1, \ldots, g_T$ generated according to samples $z_1, \ldots, z_T \sim p$ we upper bounded

$$\mathbb{E}_{z_1,\ldots,z_T}[f_p(w) - \min_w f_p(w)] \equiv \mathbb{E}_{\mathbb{P}^T}[f_p(w) - \min_w f_p(w)] \leq \frac{\|w_1 - w^*\|(G+\sigma)}{\sqrt{T}} \tag{2}$$

  where $\mathbb{P}^T$ is the joint distribution of observations $g_1, \ldots, g_T$
- How good is the suboptimality gap $\mathbb{E}_{\mathbb{P}^T}[f_p(w) - \min_w f_p(w)]$? depends on p which is unknown.
- The above notion of "risk" cannot be minimized since it depends on the unknown parameter p. Thus, to talk about the optimality, in general additional criteria are required
- One such criteria is minimax risk: achieves the smallest maximum risk among all alternative distributions belonging to some distribution set D
- Let $\mathbb{P}^T$ be the joint distribution of observations $g_1, \ldots, g_T$:

$$\mathcal{R}_T^D(w) = \max_{p \in D} \ \mathbb{E}_{\mathbb{P}^T}[f_p(w) - \min_w f_p(w)], \qquad w_{opt} = \arg\min_w \mathcal{R}_T^D(w) \tag{3}$$

Procedure to Establish the Lower Bound



- Our main strategy will involve lower bounding the risk step-by-step and constructing a hard example
- We reduce the optimization for this hard example to a statistical decision making (hypothesis test) problem
- We use tools from information theory (e.g. Le Cam's, Fano's, and Assoud's methods) to study the probability of failing the test
- We assume w.l.o.g. D is a finite set $D = \{p_1, p_{-1}\}$ and let π be a distribution over D such that $\pi(p_1) = \pi(p_{-1}) = 0.5$. Then, since max is larger than average

$$\begin{aligned}\mathcal{R}_T^D(w) &= \max_{p_i \in D} \ \mathbb{E}_{\mathbb{P}_i^T}[f_{p_i}(w) - \min_w f_{p_i}(w)] \geq \mathbb{E}_{p_i \sim \pi}\left[\mathbb{E}_{\mathbb{P}_i^T}[f_{p_i}(w) - \min_w f_{p_i}(w)]\right] \\ &= \pi(p_1)\mathbb{E}_{\mathbb{P}_1^T}[f_{p_1}(w) - \min_w f_{p_1}(w)] + \pi(p_{-1})\mathbb{E}_{\mathbb{P}_{-1}^T}[f_{p_{-1}}(w) - \min_w f_{p_{-1}}(w)]\end{aligned} \tag{4}$$

where $\mathbb{P}_i^T$ is the joint distribution of observations (stochastic gradients) when $z \sim p_i$

Binary Hypothesis Testing



- We will construct a test based on the optimization performance $f_{p_i}(w) - \min_w f_{p_i}(w)$
- In hypothesis testing, the goal is to determine which of the two alternative is correct based on a number of observations.
- Nature chooses a distribution $p_i \in D$ and as a result a loss function $f_{p_i}(w)$ (in our case $D = \{p_1, p_{-1}\}$)
- Based on this choice, we observe stochastic gradients $g_1, \ldots, g_T$ for $f_{p_i}(w)$ with joint distribution $\mathbb{P}_i^T$
- Goal is to correctly identify the choice made by nature, i.e., from $g_1, \ldots, g_T$ decide which distribution/loss was chosen.
- Typically, the probability of making mistake is lower bounded as the distributions, despite being different, are similar, so the observations could lead to a mistake
- The implication for us will be a lower bound on the convergence rate

Construction of Loss functions

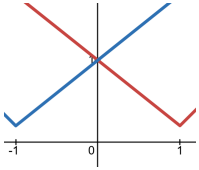
- Let $\ell(w, z) = G|w - zD|$ for $z \in \{1, -1\}$. Note the function is convex and G-Lipschitz. Consider two binary distribution on z

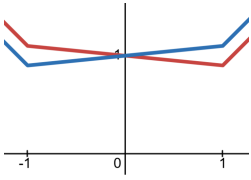
$$p_1(z = i) = \frac{1 + i\delta}{2}, \qquad p_{-1}(z = i) = \frac{1 - i\delta}{2}, \qquad i \in \{1, -1\} \tag{5}$$

giving rise to the two functions

$$f_{p_i}(w) = \mathbb{E}_{z \sim p_i}[\ell(w, z)] = \frac{1 + i\delta}{2} G|w - D| + \frac{1 - i\delta}{2} G|w + D|, \qquad i \in \{1, -1\} \tag{6}$$

- Both functions are convex and G-Lipschitz and also $w_i^* = iD$ implying $|w_i^*| = D$
- Functions becomes harder to distinguish as δ gets smaller





Construction of Loss functions (Cont'd)



- We consider the natural choice $g_t = G \cdot \text{sign}(w_t - z_t D)$ for the stochastic gradients by noting $\text{sign}(w - D)$ is a (sub)gradient of $|w - D|$
- It is evident that g_t is an unbiased estimator of $\nabla f_i(w_t)$ under $z_t \sim p_i$ and $\text{var}(g_t) \leq \mathbb{E}\|g_t\|^2 \leq G^2$
- The idea is that since the functions are so similar for small δ, by observing g_t we might mistakenly think they are the stochastic gradient for the other function.
- Despite the similarity of functions and stochastic gradients, the functions are separated in the sense that if

$$f_{p_i}(\hat{w}) \leq f_{p_i}(w_i^*) + 2GD\delta \quad \Rightarrow \quad f_{p_{-i}}(\hat{w}) \leq f_{p_{-i}}(w_{-i}^*) + 2GD\delta, \qquad i \in \{1, -1\} \tag{7}$$

- We will use this separation property to reduce optimization to statistical test



- We observe the sequence of stochastic gradients and use them in our favorite algorithm and we output p_i as the candidate for the initial choice of nature if $f_{p_i}(\hat{w}) \leq f_{p_i}(w_i^*) + 2GD\delta$
- Let us denote this test by $\mathrm{Te}(g_1, \ldots, g_T) = i$
- Recall we showed

$$\mathcal{R}_T^D(w) = \max_{p_i \in D} \ \mathbb{E}_{\mathbb{P}_i^T}[f_{p_i}(w) - \min_w f_{p_i}(w)] \geq \mathbb{E}_{p_i \sim \pi}\left[\mathbb{E}_{\mathbb{P}_i^T}[f_{p_i}(w) - \min_w f_{p_i}(w)]\right] \tag{8}$$

- By Markov inequality, i.e. $\Pr(X \geq \lambda) \leq \mathbb{E}[X]\lambda^{-1}$ for $X, \lambda > 0$

$$\mathcal{R}_T^D(w) \geq \mathbb{E}_{p_i \sim \pi}\left[\mathbb{P}_i^T(f_{p_i}(w) - \min_w f_{p_i}(w) \geq 2GD\delta)\right] \cdot 2GD\delta \tag{9}$$

Error of Hypothesis Test and Optimization (Cont'd)



- Note $\mathbb{P}_i^T(f_p(w) - \min_w f_p(w) \geq 2GD\delta)$ is precisely the probability of making an error (failing the test) via our test $\mathsf{Te}(g_1, \ldots, g_T)$ when $g_1, \ldots, g_T$ are generated according to p_i.
- Total probability of error:

$$\begin{aligned}\mathbb{Q}(\mathsf{Te}(g_1, \ldots, g_T) \neq i) &= \pi(p_1) \cdot \mathbb{P}_1^T(\mathsf{Te}(g_1, \ldots, g_T) \neq 1) \\ &\quad + \pi(p_{-1}) \cdot \mathbb{P}_{-1}^T(\mathsf{Te}(g_1, \ldots, g_T) \neq -1) \\ &= \mathbb{E}_{p_i \sim \pi}\Big[\mathbb{P}_i^T(\mathsf{Te}(g_1, \ldots, g_T) \neq i)\Big]\end{aligned} \tag{10}$$

  where $\mathbb{Q}$ is the joint distribution over the random index $i \in \{-1, 1\}$ and the observed gradients $g_1, \ldots, g_T$
- Therefore, we have effectively shown

$$\mathcal{R}_T^D(w) \geq \mathbb{Q}(\mathsf{Te}(g_1, \ldots, g_T) \neq i) \cdot 2GD\delta \geq \min_{\tilde{\mathsf{Te}} \in \text{ set of all Tests}} \mathbb{Q}(\tilde{\mathsf{Te}}(g_1, \ldots, g_T) \neq i) \cdot 2GD\delta \tag{11}$$

Bounding the Error Probability



- Recall we have set $\pi(p_1) = \pi(p_{-1}) = 0.5$.
- Also note $\mathbb{P}_i^T$ are different probability measures defined on the same sample space
- Further the events corresponding to $\tilde{\text{Te}}(g_1, \ldots, g_T) \neq 1$ and $\tilde{\text{Te}}(g_1, \ldots, g_T) \neq -1$ are complement: any testing procedure maps one region of the sample space A, to 1 and the complement (A^c) to -1. Thus,

$$
\begin{aligned}
&\min_{\tilde{\text{Te}} \in \text{ set of all Tests}} \mathbb{Q}(\tilde{\text{Te}}(g_1, \ldots, g_T) \neq i) \\
&= \frac{1}{2} \min_{\tilde{\text{Te}} \in \text{ set of all Tests}} \left(\mathbb{P}_1^T(\tilde{\text{Te}}(g_1, \ldots, g_T) \neq 1) + \mathbb{P}_{-1}^T(\tilde{\text{Te}}(g_1, \ldots, g_T) \neq -1) \right) \\
&\equiv \min_A \frac{1}{2} \left(\mathbb{P}_1^T(A^c) + \mathbb{P}_{-1}^T(A) \right) \equiv \min_A \frac{1}{2} \left(1 - \mathbb{P}_1^T(A) + \mathbb{P}_{-1}^T(A) \right) \\
&\equiv \frac{1}{2} \left(1 - \max_A \{ \mathbb{P}_1^T(A) - \mathbb{P}_{-1}^T(A) \} \right) \equiv \frac{1}{2} (1 - \| \mathbb{P}_1^T - \mathbb{P}_{-1}^T \|_{TV})
\end{aligned}
\tag{12}
$$

Bounding the Minimax Rate with KL



- Thus, we have established a lower bound on the risk in terms of the separation of our functions and the joint distributions:

$$\mathcal{R}_T^D(w) \geq GD\delta(1 - \|\mathbb{P}_1^T - \mathbb{P}_{-1}^T\|_{TV}) \tag{13}$$

- One then typically upper bound the TV distance with KL divergence via

$$\text{Pinsker:} \quad \|\mathbb{P}_1^T - \mathbb{P}_{-1}^T\|_{TV} \leq \sqrt{\frac{1}{2} D_{kl}(\mathbb{P}_1^T \| \mathbb{P}_{-1}^T)} \tag{14}$$

$$\text{Bertagnolle-Huber:} \quad \|\mathbb{P}_1^T - \mathbb{P}_{-1}^T\|_{TV} \leq \sqrt{1 - \exp(-D_{kl}(\mathbb{P}_1^T \| \mathbb{P}_{-1}^T))} \tag{15}$$

Tensorization of KL Lemma

Let $\mathbb{P}_i^{[t]}$ denote the conditional distribution of g_t given $g_1, \ldots, g_{t-1}$ when $z \sim p_i$. Then,

$$D_{kl}(\mathbb{P}_1^T \| \mathbb{P}_{-1}^T) = \sum_{t=1}^{T} \mathbb{E}_{\mathbb{P}_1^{t-1}}[D_{kl}(\mathbb{P}_1^{[t]} \| \mathbb{P}_{-1}^{[t]})] \leq T \max_{t=1,\ldots,T} D_{kl}(\mathbb{P}_1^{[t]} \| \mathbb{P}_{-1}^{[t]}) \tag{16}$$

Bounding the Minimax Rate with KL (Cont'd)



- Thus, using KL tensorization our lower bound reduces to

$$\mathcal{R}_T^D(w) \geq GD\delta(1 - \sqrt{0.5T \max_{t=1,\ldots,T} D_{kl}(\mathbb{P}_1^{[t]} \| \mathbb{P}_{-1}^{[t]})}) \tag{17}$$

- Note that for any t, $\mathbb{P}_i^{[t]}$ is basically a Bernoulli distribution:

$$\mathbb{P}_1^{[t]}(z = i) = \frac{1 + i\delta}{2}, \qquad \mathbb{P}_{-1}^{[t]}(z = i) = \frac{1 - i\delta}{2}, \qquad i \in \{-1, 1\} \tag{18}$$

- So, we need to just calculate the KL between two slightly different Bernoulli distributions:

$$D_{kl}(\mathbb{P}_1^{[t]} \| \mathbb{P}_{-1}^{[t]}) = \delta \log \frac{1 + \delta}{1 - \delta} \leq c_1 \delta^2 \tag{19}$$

for some $c > 1$ and $\delta < 1$ (using Taylor approximation of log). Thus,

$$\mathcal{R}_T^D(w) \geq GD\delta(1 - c_2\delta\sqrt{T}) \geq c_3 \frac{DG}{\sqrt{T}} \tag{20}$$

by setting $\delta = c_4/\sqrt{T}$ for some $c_2, c_3, c_4 > 0$